

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 8 — Employee Performance Analysis

Aim

To analyze employee performance data using **line chart, bar chart, table, and Tableau dashboard** for interactive exploration of employee performance.

Dataset

Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Q1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R Program

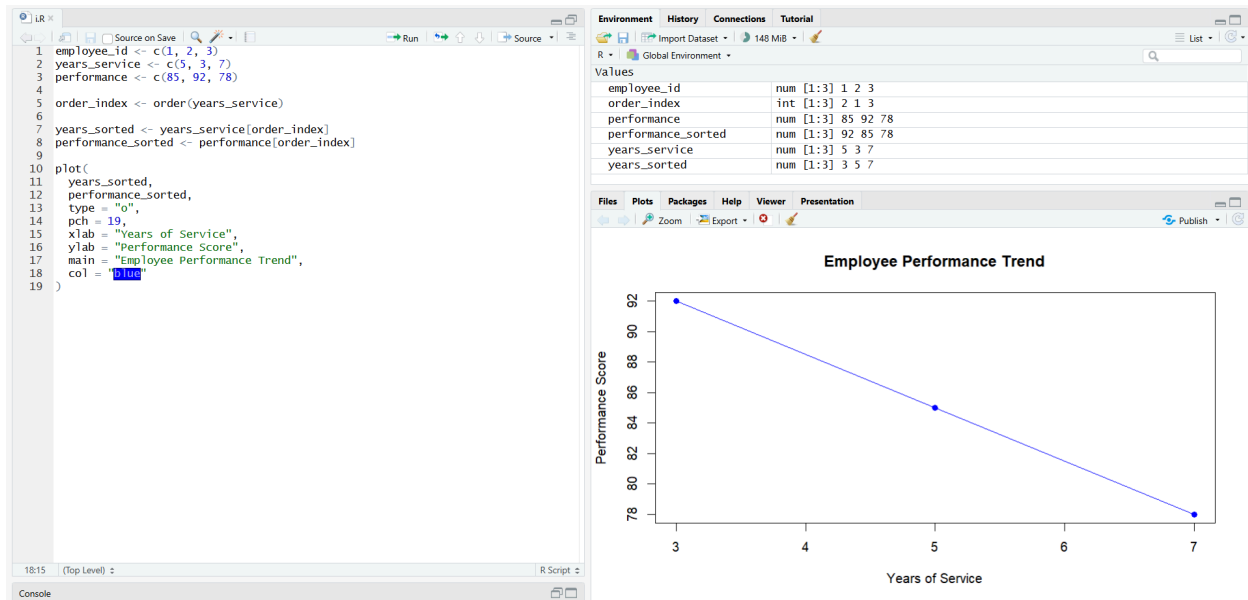
```
employee_id <- c(1, 2, 3)
years_service <- c(5, 3, 7)
performance <- c(85, 92, 78)

order_index <- order(years_service)

years_sorted <- years_service[order_index]
performance_sorted <- performance[order_index]

plot(
  years_sorted,
  performance_sorted,
  type = "o",
  pch = 19,
  xlab = "Years of Service",
  ylab = "Performance Score",
  main = "Employee Performance Trend",
  col = "blue"
)
```

Output



Observation

- Employee with **3 years of service** has a performance score of **92**.
- Employee with **5 years of service** has a score of **85**.
- Employee with **7 years of service** has a score of **78**.
- In this small dataset, the performance score decreases as years of service increase.

Q2. Generate a bar chart showing the distribution of employees across different departments.

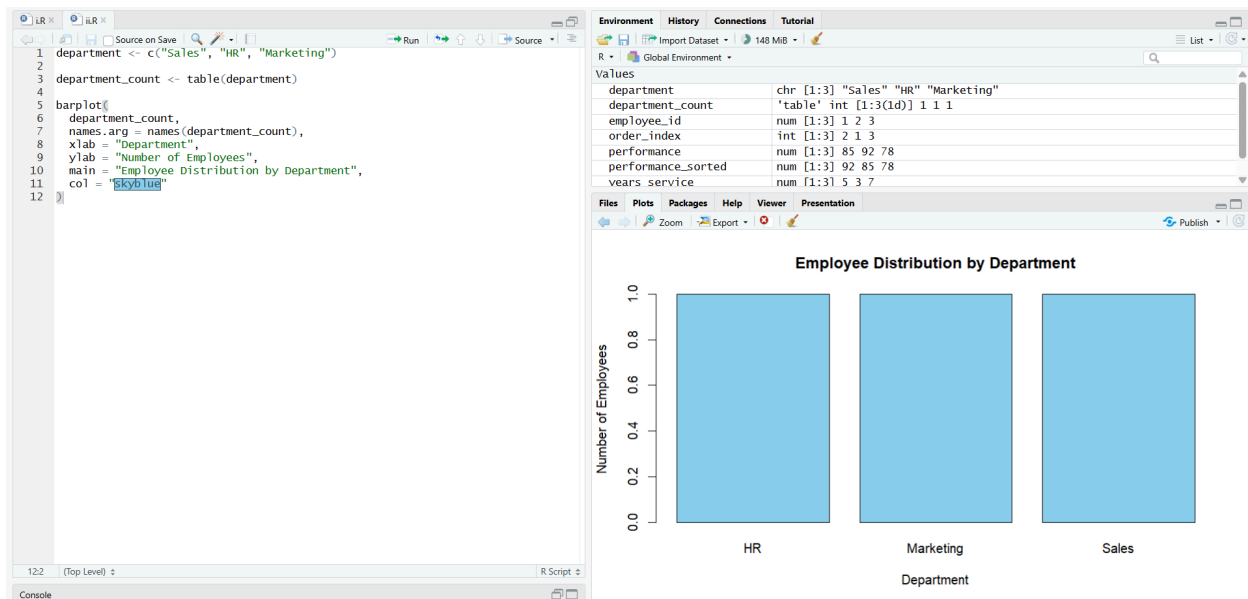
R Program

```
department <- c("Sales", "HR", "Marketing")

department_count <- table(department)

barplot(
  department_count,
  names.arg = names(department_count),
  xlab = "Department",
  ylab = "Number of Employees",
  main = "Employee Distribution by Department",
  col = "skyblue"
)
```

Output

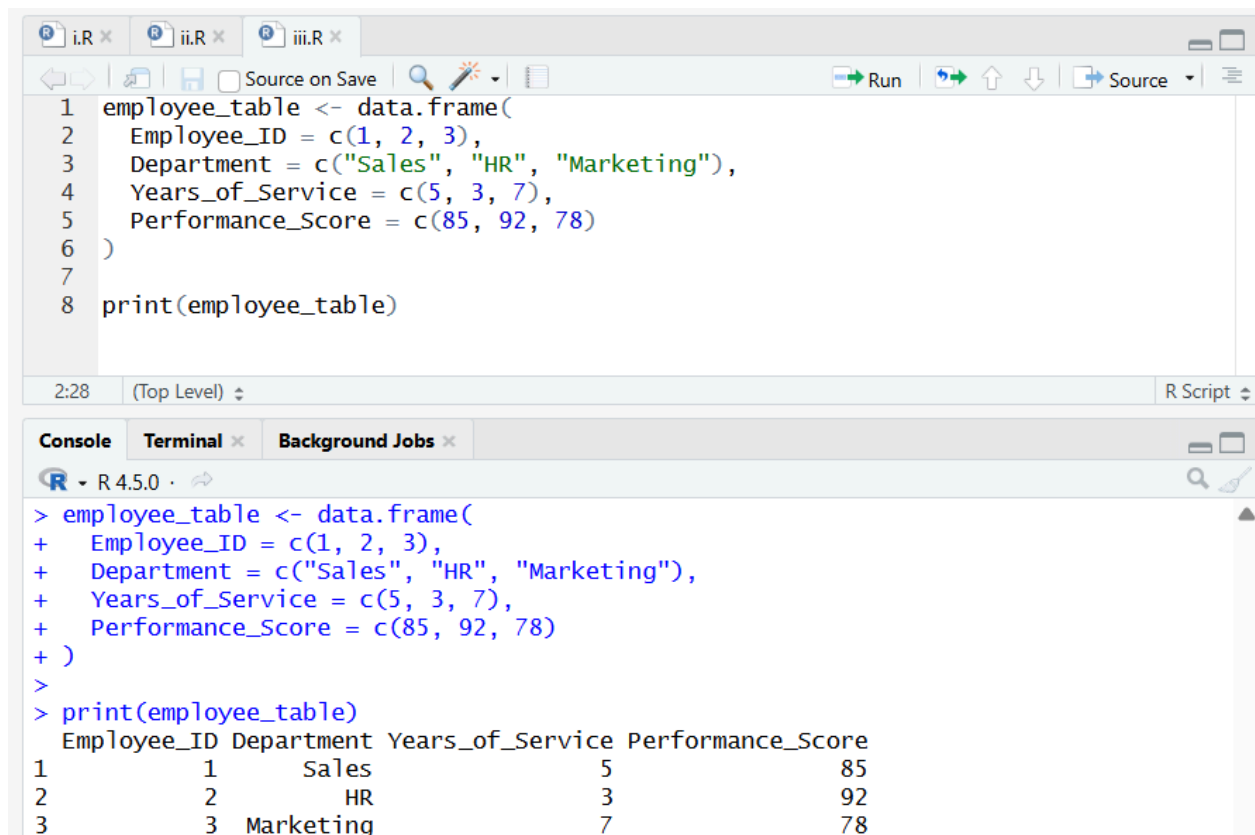


Q3. Build a table to display the performance data for each employee.

R Program

```
employee_table <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
  
print(employee_table)
```

Output



The screenshot shows the R Studio environment. The script editor contains the R code to create a data frame and print it. The console shows the execution of the code, resulting in a table with 3 rows and 4 columns: Employee_ID, Department, Years_of_Service, and Performance_Score.

```
1 employee_table <- data.frame(  
2   Employee_ID = c(1, 2, 3),  
3   Department = c("Sales", "HR", "Marketing"),  
4   Years_of_Service = c(5, 3, 7),  
5   Performance_Score = c(85, 92, 78)  
6 )  
7  
8 print(employee_table)
```

```
> employee_table <- data.frame(  
+   Employee_ID = c(1, 2, 3),  
+   Department = c("Sales", "HR", "Marketing"),  
+   Years_of_Service = c(5, 3, 7),  
+   Performance_Score = c(85, 92, 78)  
+ )  
>  
> print(employee_table)  
  Employee_ID Department Years_of_Service Performance_Score  
1           1     Sales              5              85  
2           2        HR              3              92  
3           3  Marketing              7              78
```

Observation

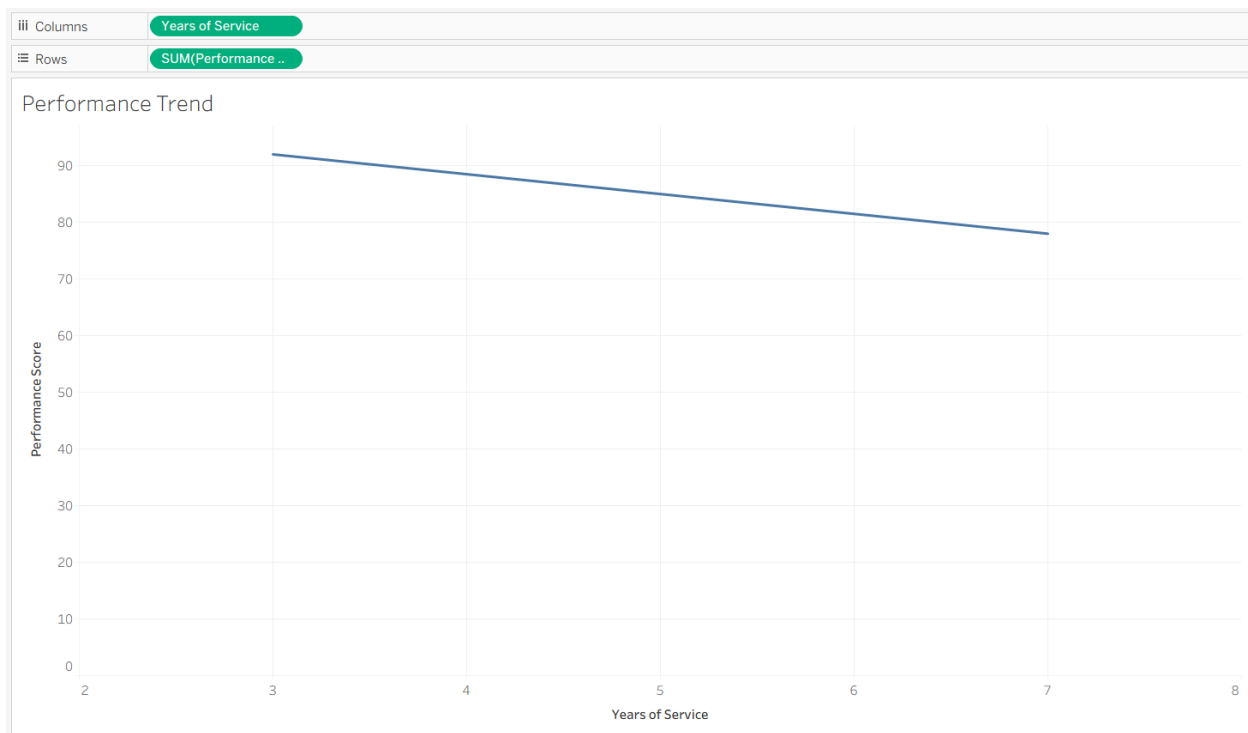
The table successfully displays the complete performance information of all three employees.

Q4. Develop a Tableau dashboard combining the line chart, bar chart, and table.

Tableau Procedure

1. Open **Tableau**.
2. Import the employee performance dataset.
3. Create a worksheet for the **line chart**:
 - Drag **Years of Service** to Columns.
 - Drag **Performance Score** to Rows.
 - Select **Line** from the Marks card.
4. Create a worksheet for the **bar chart**:
 - Drag **Department** to Columns.
 - Drag **Employee ID** to Rows.
 - Use **COUNT(Employee ID)** to show the number of employees.
 - Select **Bar**.
5. Create a worksheet for the **employee performance table**:
 - Add Employee ID, Department, Years of Service, and Performance Score.
6. Select **New Dashboard**.
7. Add the **line chart**, **bar chart**, and **performance table** to the dashboard.
8. Add suitable filters and formatting for interactive exploration.

Output:



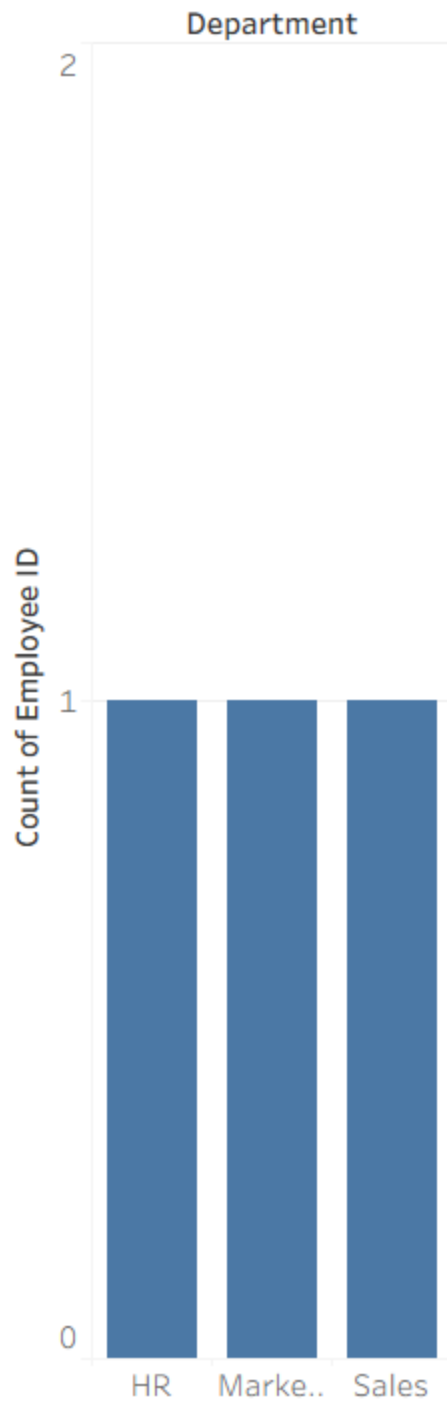
iii Columns

Department

Rows

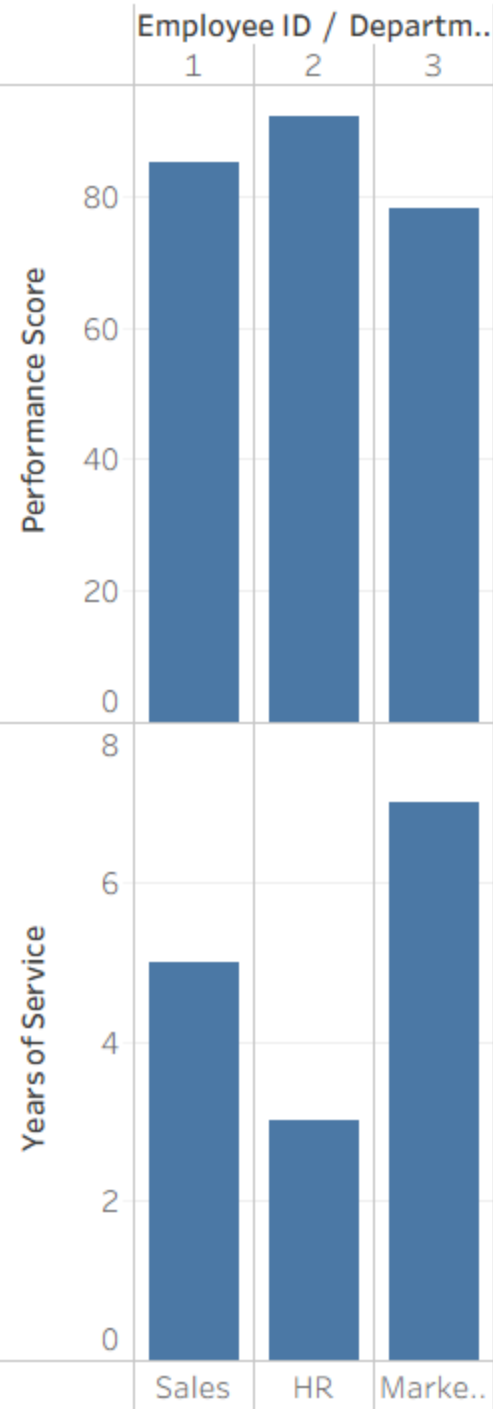
CNT(Employee ID)

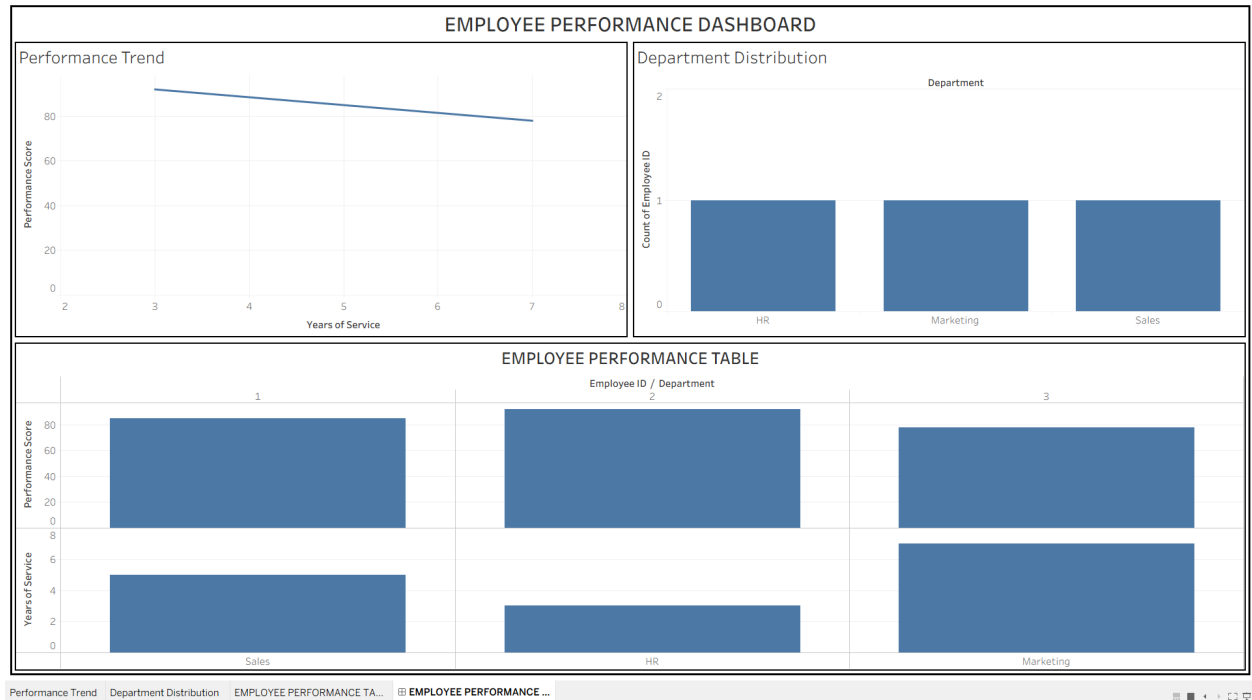
Department Distribution



iii Columns	Employee ID	Department
iii Rows	SUM(Performance ..	SUM(Years of Servi..

EMPLOYEE PERFORMANCE TABLE





Procedure

1. Open **RStudio** and create a new R Script.
2. Enter the given employee performance dataset.
3. Use the `plot()` function to create a **line chart** of performance score against years of service.
4. Use the `table()` and `barplot()` functions to visualize the **distribution of employees across departments**.
5. Use `data.frame()` to create a **table containing employee performance information**.
6. Open **Tableau** and import the employee performance dataset.
7. Create the line chart, department bar chart, and employee performance table in separate worksheets.
8. Create a new Tableau dashboard and combine all three visualizations.
9. Add suitable titles, labels, filters, and formatting for interactive exploration.

Result

The employee performance data was successfully analyzed using **R and Tableau**. A line chart was created to visualize the performance trend, a bar chart was created to show employee distribution across departments, and a table was created to display individual employee performance data. These visualizations were successfully combined into an **interactive Tableau dashboard**.